

Project: COVID-19 Vaccine Accountability Dashboard (Ghana EPI)

Background

Ghana's Expanded Programme on Immunization (EPI) is responsible for COVID-19 vaccine accountability and monitoring. This involves tracking the usage of COVID-19 vaccines, ensuring quality, and evaluating program performance to maximize impact and ensure proper resource allocation. This process includes monitoring the COVID-19 vaccine supply chains, tracking immunization coverage, and assessing the effectiveness of immunization strategies. Key aspects of this initiative include monitoring vaccine wastage, managing the cold chain, and addressing adverse events following immunization (AEFI).

The objective of this dashboard project was to enhance oversight of COVID-19 vaccine accountability by reporting and analysis:

- Unusable vaccines (damaged, broken vials, improperly handled)
- Wasted vaccines (open vial wastage, unused doses after sessions)
- Expired vaccines (past manufacturer expiry date)

This was critical for:

- Ensuring proper waste classification and disposal (e.g., incineration)
- Supporting cold chain compliance and quality control
- Aiding donor transparency and reporting
- Informing redistribution and resupply decisions

Approach

1. Data Collection & Sources

- Data was submitted weekly from all COVID vaccination centres via Kobo Toolbox. Kobo Toolbox data connected to Power BI via Kobo API.

Key fields captured:

- Region, District, Facility Name
- Vaccine Type (e.g., AstraZeneca, Moderna, Pfizer, J&J,Sputnik_V)
- Batch Number
- Number of vials: Used, Usable, Missing, Expired, Broken, Spoilt
- Reason for wastage
- Date of reporting

2. Data Modelling

- Data extracted from Kobo Toolbox.
- Cleaned and structured using Power Query
- Model created:
 - Fact_vaccines – contains all vaccine records
 - Dim_Facility, Dim_Location, Dim_VaccineList
- Created calculated measures using DAX (e.g., ttl_Broken_Vials, ttl_Used_Vials, ttl_Expired_Vials etc.)

3. Dashboard Features

- **KPIs:** Vaccine Districts, Vaccine Centres, Issued_Vials, Used_Vials, Usable_Vials, Expired_Vials, Missing_Vials, Broken_Vials, Spoilt_Vials.
- **Visuals:**
 - Table showing name of vaccine and the quantity issued, expired, missing, broken, spoilt, all in vials.
 - Funnel showing the percentages of usable, used, expired, missing, broken, and spoilt vials.
 - Clustered column chart showing total vials issued by vaccine type versus vials usable, used, missing, expired, broken, and spoilt.
 - Tables showing vaccine counts in vials versus in doses for usable, used, expired, missing, broken, and spoilt.

- Clustered column chart showing percentage of issued vials by region, percentage of usable vials by region, percentage of expired vials by region, percentage of used vials by region, percentage of broken vials by region, percentage of spoilt vials by region, and percentage of missing vials by region.
- **Interactivity:**
 - **Slicer:** Filter by Vaccine Type, Filter by Region & District.

4. **Deployment**

- Published to Power BI Service.
- Scheduled data refresh weekly.
- Shared with Vaccine Program Leads and District /Regional Officers via Power BI Service

Impact

- Centralized real-time reporting of unusable and expired vaccines nationwide.
- Enabled timely planning of incineration logistics by regional cold stores.
- Reduced reporting delays compared to legacy manual summaries.
- Supported accountability in reporting to COVAX and development partners.